TEXORA · LECTURE SHEET

Metals and Alloys

Iron and its family, then the rest: the iron–carbon system in outline, steel grades and cast irons, stainless steel's passive film, the strengthening mechanisms every alloy exploits, and the great non-ferrous systems — aluminium, copper and titanium.

ME-1207 · ENGINEERING MATERIALS · TOPIC 3 · LEVEL 1 TERM 2

1. Why alloy at all

Pure metals are soft, easily corroded and hard to tailor. **Alloying** — deliberately dissolving other elements — buys strength, hardness, corrosion resistance and heat response. Four strengthening mechanisms recur everywhere; learn them once, recognise them forever:

MECHANISM	IDEA / EXAMPLE
Solid-solution strengthening	Guest atoms strain the lattice and hinder dislocations — Zn in Cu (brass outmuscles both parents)
Grain refinement (Hall–Petch)	σ_y = σ_0 + k d^(−1/2): grain boundaries block slip, so finer grains = stronger *and* tougher — the only mechanism that improves both
Precipitation (age) hardening	A shower of fine second-phase particles pins dislocations — duralumin, hardened over days at room temperature or hours in an oven
Work hardening + phase transformation	Cold work multiplies dislocations until they tangle; quenching steel traps the martensite phase — the hardest trick of all (ME-1208)

WORKED EXAMPLE — HALL–PETCH

For mild steel take σ_0 = 70 MPa, k = 0.7 MPa·m^½. At grain size d = 100 μm: d^(−1/2) = (10^{-4})^(−1/2) = 100 → σ_y = 70 + 70 = **140 MPa**. Refine to d = 10 μm: d^(−1/2) = 316 → σ_y = 70 + 221 = **291 MPa**. Tenfold finer grains, double the yield strength — free strength, bought only with processing.

2. Ferrous metals — the iron–carbon world

Carbon content draws the master line: **steels** carry up to ≈ 2% C; **cast irons** 2–4% (castable, because the extra carbon lowers the melting range). The room-temperature phases: **ferrite** (BCC α-iron,

dissolves ≤ 0.02% C — soft and ductile), **cementite** Fe_3C (6.67% C — glass-hard and brittle), and **pearlite**, the lamellar ferrite+cementite mixture that forms when austenite (FCC γ, stable hot) decomposes at the eutectoid point, 0.77% C and 727 °C. A 0.4% C steel slowly cooled is therefore about half free ferrite and half pearlite (pearlite fraction ≈ 0.4/0.77 ≈ 52%) — soft enough to machine, strong enough to work; a 0.77% steel is fully pearlite; above it, cementite networks embrittle the grain boundaries.

GRADE	CARBON	CHARACTER & USES
Low-carbon (mild) steel	< 0.25%	Ductile, weldable, cheap — sheet, wire, structures, machine frames
Medium-carbon steel	0.25–0.6%	Heat-treatable balance — shafts, gears, axles
High-carbon steel	0.6–1.4%	Hard, wear-resistant — springs, tools, rails, needles
Stainless steel	+ ≥ 10.5% Cr	A self-repairing Cr_2O_3 passive film stops rust; 18/8 grade (18% Cr, 8% Ni) is austenitic, non-magnetic, the dyehouse and food-plant standard
Grey cast iron	2–4% C, Si	Graphite flakes: superb damping and machinability, weak in tension — engine blocks, machine beds
Ductile (nodular) iron	+ Mg treatment	Graphite as spheres, not flakes: castable *and* tough — crankshafts, pipes

Alloying elements in steel read like a menu: Mn (strength, hardenability, deoxidiser), Cr (hardness, corrosion, carbides), Ni (toughness at low temperature), Mo (high-temperature strength, resists temper brittleness), V (fine grains, wear), W (hot hardness in high-speed tool steel).

3. Non-ferrous metals

- **Aluminium** (2.70 g cm^{-3}, FCC): corrosion-proof by its own oxide, highly conductive, cast and wrought families. **Duralumin** (Al + ≈ 4% Cu + Mg, Mn) age-hardens to steel-like strength at a third of the weight — the original aircraft alloy.
- **Copper** (8.96 g cm^{-3}, FCC): the electrical metal; supreme conductivity and antimicrobial surfaces. **Brass** = Cu–Zn (ductile, corrosion-resistant — fittings, zips, buttons in garments); **bronze** = Cu–Sn (hard, low-friction — bearings, bushings, bells). Bronze outlasted an entire age named after it.
- **Titanium** (4.43 g cm^{-3}): strength rivalling steel at half weight, superb corrosion resistance, costly extraction — aerospace, implants, chemical plant.
- **Magnesium** (1.74 g cm^{-3}): lightest structural metal, HCP and ill-mannered cold (Topic 1), flammable in chips — laptop cases, gearbox housings.

4. Key points

- Four strengthening levers: solid solution, grain refinement (Hall–Petch), precipitation, work/transformation hardening.
- Steel < 2% C, cast iron 2–4%; ferrite soft, cementite hard, pearlite the eutectoid mixture at 0.77% C / 727 °C.
- Stainless = ≥ 10.5% Cr and a self-healing oxide. Grey iron's flakes damp vibration; nodular iron's spheres restore toughness.
- Aluminium alloys sell lightness; copper alloys sell conductivity and corrosion behaviour; titanium sells strength-to-weight at a price.

5. Practice

Answer in your own words; brief model answers follow.

1. Using σ_0 = 70 MPa and k = 0.7 MPa·m^½, find the yield strength of a steel with grain size 25 µm.
2. A 0.4% carbon steel is cooled slowly from the austenite region. Describe its room-temperature microstructure and estimate the proportions.
3. Why does adding 12% chromium turn steel "stainless", and why does the protection fail in a chloride-rich dye bath at welds that were not properly treated?
4. Grey cast iron machine beds damp vibration better than steel beds of the same casting. Explain, using graphite morphology — and state what the magnesium-treated version changes and gains.
5. Duralumin left for some days after quenching becomes markedly harder without further heating. Name the mechanism and the microstructural event.
6. A garment factory needs corrosion-proof zip sliders and snap buttons, easily cast and polished. Choose between brass, bronze, mild steel and duralumin, and justify.

Answer key

1. d = 25 µm = 2.5 × 10⁻⁵ m; d^(−1/2) = √(1/2.5 × 10⁻⁵) = √40,000 = 200. σ_y = 70 + 0.7 × 200 = 210 MPa.
2. Proeutectoid ferrite plus pearlite colonies: pearlite fraction ≈ 0.4/0.77 ≈ 52%, ferrite ≈ 48%. The ferrite formed first as the cooling crossed into the two-phase field; the remaining austenite, enriched to 0.77% C, transformed to pearlite at 727 °C.
3. At ≥ ≈ 10.5% Cr the surface grows a continuous, adherent Cr_2O_3 film only nanometres thick that blocks oxygen and re-forms instantly when scratched — passivity. Chlorides locally break the film (pitting/crevice attack), and welding can precipitate chromium carbides at grain boundaries, draining

Cr from the zones beside them (sensitisation); those zones then corrode intergranularly unless low-carbon or stabilised grades, or post-weld treatment, are used.

4. In grey iron the graphite exists as sharp flakes dispersed through the matrix. The flake–matrix interfaces and the soft graphite itself absorb vibrational energy (internal friction), so the bed rings less. The same flakes, though, are internal cracks: they slash tensile strength and ductility. Treating the melt with magnesium makes graphite grow as spheres instead of flakes — nodular/ductile iron — removing the crack tips and recovering much of steel's toughness while keeping castability and some damping.
5. Precipitation (age) hardening. Quenching traps a supersaturated solid solution; at room temperature copper-rich clusters and fine precipitates (GP zones, then θ″/θ′ phases) form slowly and strain the lattice, pinning dislocations — strength climbs over days (natural ageing), or in hours at ≈ 170 °C (artificial ageing).
6. Brass. Copper–zinc casts and polishes beautifully, shrugs off sweat and washing corrosion (unlike mild steel, which rusts, and duralumin, which tarnishes and is a structural alloy, not a trim alloy), machines crisply for sliders and snaps, and its colour sells as premium trim. Bronze would work but is harder, dearer and chosen for bearings, not buttons.
